

Clinical Trial Matching Based On Multi-agent debate

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Background

- Traditionally, patient–trial matching relies on labor-intensive manual review of patient records against trial criteria.
 - inefficient
 - hard to scale
 - missed enrollment opportunities
- How to recruit eligible patients in a timely and efficient manner
- Four aspects:
 - Candidate trial retrieval
 - Per-criterion eligibility assessment
 - Trial-level ranking
 - Human-in-the-loop closure
 - what information is missing to guide additional tests

Motivation

- LLM-as-judge
 - We use an LLM as a judge to perform a structured review of the system output
 - It is not meant to generate the final decision, instead, it answers questions such as whether it is good.
 - Enhance interpretability by turning the judge's outputs into auditable error-analysis artifacts
- In-context learning
 - Use the judge's feedback as additional context in the next iteration (feedback-as-context), guiding the agent to revise its reasoning and produce a more reasonable answer

Literature Review

- Rule-Based Methods: These methods, such as combining [Criteria2Query](#) with NLP, generally suffer from poor generalizability and require extensive maintenance.
- Deep learning approaches mitigate some of these limitations by improving feature extraction and modeling the inherent sequential dependencies in clinical text. However, they typically rely on large-scale, high-quality annotated datasets, which are scarce in the biomedical domain.

Literature Review

- LLM-Based Methods: Large Language Models often experience hallucination, which compromises accuracy and interpretability ([TrialGPT](#)). Furthermore, fine-tuning an open-source LLM for trial matching requires extensive computational resources, limiting its practicality ([TrialLlama](#)).
 - For TrialGPT, key drawbacks include a lack of transparency and interpretability, as well as susceptibility to hallucinations.

Methodology

- For patient record:
 - Split the patient record into a list of complete sentences to serve as the evidence pool.
- For clinical trial:
 - Extract the inclusion and exclusion criteria from each clinical trial, then segment them into a list of individual criteria.

"text": "A 25-year-old woman presents to the clinic complaining of prolonged fatigue. She denies difficulty sleeping and sleeps an average of 8 hours a night. She also notes hair loss, a change in her voice and weight gain during the previous 6 months. She complains of cold intolerance. On examination she has a prominent, soft, uniform anterior cervical mass at the midline."

Methodology

- Proposer:
 - Evaluate the criteria using CoT to generate a label, and provide evidence and rationale.
 - Present the result to the skeptic
 - Receive the skeptic's feedback, and based on this feedback, consider whether the original decision needs further elaboration or if the label needs to be modified.
- Skeptic:
 - The skeptic accepts the proposer's viewpoint and audits the proposer's thought process step-by-step based on common errors.
 - Reasoning based on imagination
 - Missing key elements of the clause
 -
 - The skeptic is only responsible for suggesting directions for improvement and does not express their own opinion.

Methodology

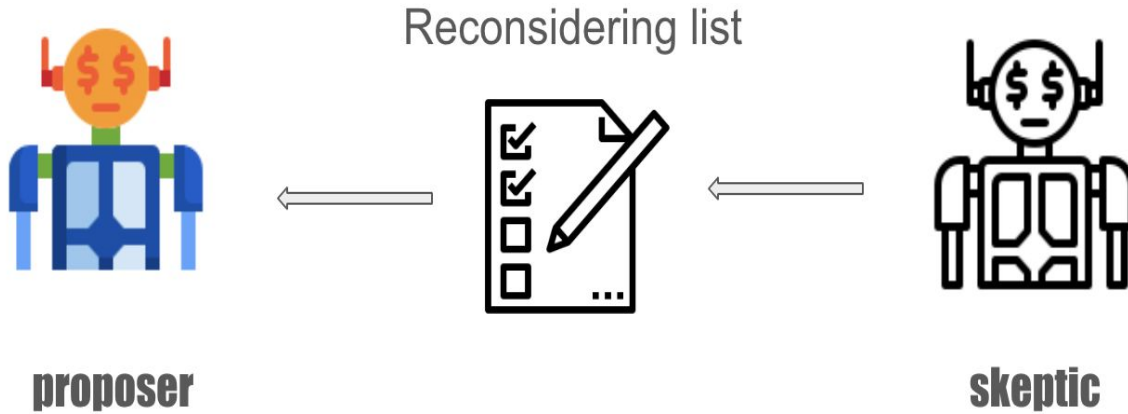


proposer



skeptic

Methodology



Methodology



proposer

Criteria: inclusion 1
Skeptic feedback: xxxxx
**Should I change label or
improve original reason?**
Evidence: xxxxx
Label: include
Reason: xxxx



skeptic

Methodology

- Proposer Agent
 - Within the scope of the entire evidence pool, produce a structured decision record for this criterion.
 - Only use evidence from the evidence pool
 - Perform CoT-style reasoning, but the final output must contain only the JSON fields.

Methodology

- Inclusion:
 - Included : there is supporting evidence
 - Not included : there is contradictory evidence
 - Otherwise : not enough information
- Exclusion
 - Exclude : there is evidence that the exclusion condition holds.
 - Not exclude : there is evidence that the exclusion condition does not hold
 - Otherwise : not enough information

For any non-NEI label, the following must hold :

- evidence is non-empty
- extractions contains at least one fact that either supports or contradicts the criterion.

Methodology

Hallucinated clinical facts: Filled up conclusion with world knowledge

Negation misinterpretation: Misreading expressions like “denies / no history of / negative for” as affirmative evidence, or flipping the polarity in the opposite direction.

Temporal and numeric reasoning errors: Mishandling operators such as “ $\geq$ / $>$ / $<$ / $\leq$ ” and time windows, leading to incorrect judgments

Criterion hallucination: Arbitrarily adding, deleting, or modifying conditions when paraphrasing the inclusion/exclusion criteria.

Evidence mismatch: Selecting sentences that are irrelevant to the focus concept.

Entity confusion: Mixing attributes from other patients or other clinical trials into the current decision.

Inconsistent label and rationale

Methodology

- Skeptic Agent
 - Ask the LLM to rely strictly on the cited sentences (evidence sentences) when checking “whether it introduces world knowledge” or other references.
 - Parse the LLM feedback: if it reports critical issues (missing citations), treat that as a revision requirement.

Methodology

- Skeptic Agent
 - Not responsible for assigning a new label or deciding include/exclude in place of the Proposer.
 - Only audits whether the Proposer's evidence, extractions, and rationale are self-consistent, sufficient
 - Output:

issues	missing_citation/invalid_sent_id.....
warnings	Severity level
support_status	support/insufficient/conflict
short_explanation	reason

Methodology

Proposer Agent (Revise)

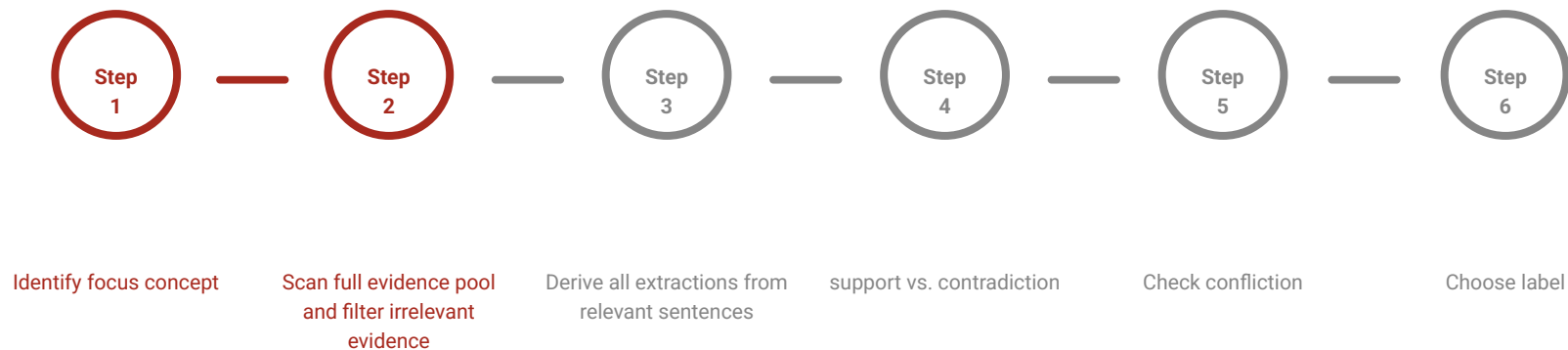
- Input: last round decision + feedback from skeptic agent
- Output: new decision
- After receiving feedback from the Skeptic agent, the Proposer will generate a new decision trace, so that the final outcome is more interpretable.

Methodology

For NEI cases, it is necessary to know exactly which pieces of patient information are missing in order to determine whether a given criterion should be considered included or excluded.

- Add a new module to the Proposer that is responsible for generating an `missing_info` field
- Enhance the interpretability of the overall system

Decision Trace



Methodology

```
{
  "clause_id": "inc_01",
  "criterion_type": "inclusion",
  "reasoning_trace": [
    {
      "step": "Identify focus concept",
      "detail": "The focus concept is 'Body mass index more than or equal to 25 kg/m2'."
    },
    {
      "step": "Scan full pool and filter irrelevant sentences",
      "detail": "Sentences [0] through [6] are not relevant as they do not mention the body mass index. Sentence [7] is also irrelevant."
    },
    {
      "step": "Derive all extractions from relevant sentences",
      "detail": "Sentence [3] mentions 'obesity', which implies a high BMI, but does not provide an explicit value."
    },
    {
      "step": "Test support vs. contradiction",
      "detail": "There is no explicit statement about the body mass index in any of the relevant sentences. Sentence [4] denies diabetes and other conditions that could affect weight, but does not directly state a BMI value."
    },
    {
      "step": "Check conflicts",
      "detail": "No contradictions are found regarding the body mass index."
    },
    {
      "step": "Choose label per rubric",
      "detail": "Since there is no explicit statement about the body mass index, and only an implication of obesity, we cannot definitively support or contradict the criterion. Therefore, 'not enough information' is chosen."
    }
  ],
  "claim": "Body mass index more than or equal to 25 kg/m2",
  "evidence": [],
  "extractions": {},
  "missing_information": [
    "specific body mass index value"
  ],
  "rationale": "While the patient is known to be obese, there is no explicit statement about her body mass index. Therefore, we do not have enough information to determine if she meets the inclusion criterion.",
  "label": "not enough information"
}
```

Methodology

What is the optimal number of iterations between the Proposer and the Skeptic?

Two rounds!

- We found that after more than two rounds, the Proposer tends to fall into a local optimum: further iterations no longer improve the decision, but instead either repeat the same outputs or introduce new hallucinations.

Experiment Set-up

Model: qwen2.5:7b-instruct

Ground_truth : Used GPT-5 to generate ground truth

You are a clinical trial eligibility reviewer. Given a patient record and ONE criterion, assign a label using ONLY the record. Do not guess. If the record is insufficient, choose Not Enough Information (NEI). If criterion_type = inclusion: included / not included / NEI; If criterion_type = exclusion: exclude / not exclude / NEI.

Patient record: {PATIENT_RECORD}

Criterion:{CRITERION_TEXT}

Evaluation

1. Randomly select one patient and 10 clinical trials from the SIGIR dataset.
2. Collect the ground-truth eligibility labels for these 10 trials.
3. Run our pipeline to produce per-criterion predictions
4. Run TrialGPT on the same patient–trial set to obtain its trial-matching outputs
5. Evaluate and compare our method against TrialGPT using the same ground truth

Evaluation

Experiment	Accuracy(multi-agent debate)	Accuracy(TrialGPT)
A	0.9375	0.875
B	0.8813	0.8305
C	0.8857	0.7857

Rank

1. Hard filter

Any trial is immediately discarded if any exclusion criterion is labeled “excluded” or if any inclusion criterion is labeled “not included.”

2. For each trial, compute the I, S, and U

$I = \text{included_count} / \text{total_inclusion}$

$S = \text{not_excluded_count} / \text{total_exclusion}$

$U = \text{total NEI clauses}$

Rank

3. For NEI:

Compute the weight and cost for each NEI criterion.

Ask, admin	1
EHR, medication	2
Labs	3
Imaging	4
Complex pathology	5

Rank

4. Get (I, S, U, GateNEI, RiskNEI, ClarifyCost, Actionability)

$\text{GateNEI} = \sum_i \text{weight} (i \text{ belongs inclusion criteria})$

$\text{RiskNEI} = \sum_i \text{weight} (i \text{ belongs exclusion criteria})$

$\text{ClarifyCost} = \sum (\text{weight} \times \text{cost})$

$\text{Actionability} = \sum (\text{weight} \div \text{cost})$

Rank

5. Rank rule: start with **I** and compare each component in order
 - Higher **I** and **S** are preferred
 - **U** should be as small as possible, so we keep it ascending.
 - GateNEI, RiskNEI, and ClarifyCost are also better when smaller, so they stay ascending.
 - Actionability is better when larger

Limitation

1. The Qwen model's capacity is limited, making it difficult to handle large-scale data
2. It is unclear whether the resulting rankings are truly reasonable, indicating a lack of end-to-end validation.
3. Per-criterion reasoning relies on a single model.

Conclusion

- Built a Proposer/Skeptic loop that automatically parses patient notes into sentence-level evidence and trial descriptions into structured inclusion/exclusion criteria, labeling each criterion.
- Design the evaluation pipeline with per-criterion metrics and trial ranking
- In per-criterion eligibility labeling, our multi-agent debate framework outperforms TrialGPT(Trial-matching).